

Week 2) Exercise 1)

We consider random variables X, Y with a joint pdf $f(x, y) = e \cdot \sin(x+y)$ for $0 \leq x, y \leq \frac{\pi}{2}$.

We see that

(a) For $f(x, y)$ to be a pdf, it must hold that $\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} e \cdot \sin(x+y) dx dy = 1$.

But we have that

$$\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} e \cdot \sin(x+y) dx dy$$

$$= e \int_0^{\frac{\pi}{2}} \int_y^{y+\frac{\pi}{2}} \sin(u) du dy$$

Setting $u = x+y$
and using integration
by substitution

$$= e \int_0^{\frac{\pi}{2}} \left(-\cos(u) \Big|_y^{y+\frac{\pi}{2}} \right) dy$$

$$= e \int_0^{\frac{\pi}{2}} \left(-\cos\left(y+\frac{\pi}{2}\right) + \cos(y) \right) dy$$

$$= e \int_0^{\frac{\pi}{2}} \left(\sin(y) + \cos(y) \right) dy$$

$$= e \cdot \int_0^{\frac{\pi}{2}} \sin(y) dy + e \cdot \int_0^{\frac{\pi}{2}} \cos(y) dy$$

$$= e \cdot \left(-\cos(y) \Big|_0^{\frac{\pi}{2}} \right) + e \cdot \left(\sin(y) \Big|_0^{\frac{\pi}{2}} \right)$$

Recall integration by substitution

$$\int_a^b f(g(q)) \cdot g'(q) dq$$

$$= \int_{u(a)}^{u(b)} f(u) du$$

where $u = g(q)$,
 $du = g'(q) dq$

$$= c \cdot (-\cos(\frac{\pi}{2}) + \cos(0)) + c \cdot (\sinh(\frac{\pi}{2}) - \sinh(0))$$

$$= c \cdot (0 + 1) + c \cdot (1 - 0)$$

$$= 2c.$$

We deduce that c must satisfy $c = \frac{1}{2}$ for $f(x,y)$ to be a proper pdf.

(b) The joint cdf of X and Y is by definition given by

$$\int_0^y \int_0^x \frac{1}{2} \sinh(z+q) dz dq$$

$$= \frac{1}{2} \int_0^y (-\cos(u) \Big|_q^{q+x}) dq$$

$$= \frac{1}{2} \int_0^y (-\cos(q+x) + \cos(q)) dq$$

$$= \frac{1}{2} \int_0^y (-\cos(q+x)) dq + \frac{1}{2} \int_0^y (\cos(q)) dq$$

$$= \frac{1}{2} (-\sin(u) \Big|_x^{x+y}) + \frac{1}{2} (\sin(u) \Big|_0^y)$$

$$= \frac{1}{2} (\sin(x) - \sin(x+y) + \sin(y) - \sin(0))$$

$$= \frac{1}{2} (\sin(x) + \sin(y) - \sin(x+y))$$

(c) The marginal pdfs of X and Y must be given by

$$f(y) = \int_0^{\pi/2} \frac{1}{2} \sin(x+y) dx$$

$$= \frac{1}{2} (\sin(y) + \cos(y))$$

we did this in (a)!

and by symmetry

$$f(x) = \int_0^{\pi/2} \frac{1}{2} \sin(x+y) dy$$

$$= \frac{1}{2} (\sin(x) + \cos(x))$$

(d) We further have

$$E[X] \stackrel{\text{def.}}{=} \int_0^{\pi/2} x \cdot f(x) dx$$

$$= \int_0^{\pi/2} x \cdot \frac{1}{2} (\sin(x) + \cos(x)) dx$$

$$= \frac{1}{2} \int_0^{\pi/2} x \sin(x) dx + \frac{1}{2} \int_0^{\pi/2} x \cos(x) dx$$

$$= \frac{1}{2} \left((-x \cos(x)) \Big|_0^{\pi/2} + \int_0^{\pi/2} \cos(x) dx \right) + \frac{1}{2} \int_0^{\pi/2} x \cos(x) dx$$

$$\uparrow \text{ using } h(x)=x, G(x)=-\cos(x) \Rightarrow \frac{dG}{dx} = \sin(x) = g(x).$$

$$= \frac{1}{2} \left(0 + (\sin(x)) \Big|_0^{\pi/2} \right) + \frac{1}{2} \int_0^{\pi/2} x \cos(x) dx$$

$$= \frac{1}{2} (0 + 1) + \frac{1}{2} \left((x \cdot \sin(x)) \Big|_0^{\pi/2} - \int_0^{\pi/2} \sin(x) dx \right)$$

$$\uparrow \text{ using } h(x)=x, G(x)=\sin(x) \Rightarrow \frac{dG}{dx} = \cos(x)$$

Recall integration by parts
 $\int_a^b h(x)g(x) dx$
 $= [h(x)G(x)]_a^b$
 $- \int_a^b h'(x)G(x) dx$
 with $\frac{dG}{dx} = g.$

$$= \frac{1}{2} + \frac{1}{2} (x \cdot \sinh(x) \Big|_0^{\pi/2}) - \frac{1}{2} \int_0^{\pi/2} \sinh(x) dx$$

$$= \frac{1}{2} + \frac{1}{2} \left(\frac{\pi}{2} \cdot \sinh\left(\frac{\pi}{2}\right) \right) - \frac{1}{2} \int_0^{\pi/2} \sinh(x) dx$$

$$= \frac{1}{2} + \frac{\pi}{4} - \frac{1}{2} (\cosh(x) \Big|_0^{\pi/2})$$

$$= \frac{1}{2} + \frac{\pi}{4} - \frac{1}{2}$$

$$= \frac{\pi}{4}.$$

And by symmetry $E[Y] = \frac{\pi}{4}$.

Furthermore (being lazy and using a calculator), we get

$$E[X^2] \stackrel{\text{def.}}{=} \int_0^{\pi/2} x^2 f(x) dx$$

$$= \int_0^{\pi/2} x^2 \cdot \frac{1}{2} (\sinh(x) + \cosh(x)) dx$$

$$= \frac{\pi}{2} + \frac{\pi^2}{8} - 2,$$

and by symmetry $E[Y^2] = \frac{\pi}{2} + \frac{\pi^2}{8} - 2$.

We now have

$$\begin{aligned} \text{Var}(X) &= \text{Var}(Y) = E[Y^2] - E[Y]^2 \\ &= \left(\frac{\pi}{2} + \frac{\pi^2}{8} - 2 \right) - \left(\frac{\pi}{4} \right)^2 \\ &= \frac{\pi}{2} + \frac{\pi^2}{16} - 2 \end{aligned}$$

Finally, we have that

$$\begin{aligned} E[X \cdot Y] &= \int_0^{\pi/2} \int_0^{\pi/2} (x \cdot y) f(x, y) dx dy \\ &= \int_0^{\pi/2} \int_0^{\pi/2} (x \cdot y) \cdot \frac{1}{2} \sinh(x+y) dx dy \\ &= \frac{1}{2} (\pi - 2) \end{aligned}$$

So

$$\begin{aligned} \text{Cor}(X, Y) &= E[XY] - E[X]E[Y] \\ &= \frac{1}{2} (\pi - 2) - \left(\frac{\pi}{4}\right)^2 \\ &= \frac{\pi}{2} - \frac{\pi^2}{16} - 1. \end{aligned}$$

(e) To find the conditional pdf $f_Y(y|x)$, we use that

$$\begin{aligned} f_Y(y|x) &\stackrel{\text{def.}}{=} \frac{f(x, y)}{f(x)} \\ &= \frac{\frac{1}{2} \sinh(x+y)}{\frac{1}{2} (\sinh(x) + \cos(x))} \\ &= \frac{\sinh(x+y)}{\sinh(x) + \cos(x)}. \end{aligned}$$

(f) Furthermore, we have

$$\begin{aligned} P(Y > X | X = x) &= P(Y > x | X = x) \\ &= \int_x^{\pi/2} f_Y(y|x) dy. \end{aligned}$$

$$= \int_x^{\frac{\pi}{2}} \frac{\sin(x+y)}{\sin(x) + \cos(x)} dy$$

$$= \frac{1}{\sin(x) + \cos(x)} \int_x^{\frac{\pi}{2}} \sin(x+y) dy$$

$$= \frac{1}{\sin(x) + \cos(x)} \int_{2x}^{x+\frac{\pi}{2}} \sin(u) du$$

Integration by
substitution
 $u = x + y$

$$= \frac{1}{\sin(x) + \cos(x)} \left[-\cos(u) \right]_{2x}^{x+\frac{\pi}{2}}$$

$$= \frac{1}{\sin(x) + \cos(x)} \cdot \left(-\cos\left(x + \frac{\pi}{2}\right) + \cos(2x) \right)$$

$$= \frac{\sin(x) + \cos(2x)}{\sin(x) + \cos(x)}$$

Is this reasonable?

$$= P(Y > X | X=0) = \frac{0+1}{0+1} = 1$$

$$= P(Y > X | X=\frac{\pi}{2}) = \frac{1-1}{1-0} = 0$$

Seems fair enough.

Book exercise 1.1)

From (1.1) and (1.2) we have an error function $E(\underline{w}) = \frac{1}{2} \sum_{n=1}^N (y(x_n, \underline{w}) - t_n)^2$

where each (prediction) $y(x_n, \underline{w}) = \sum_{i=0}^M w_i x_n^i$.

Noting that $E(\underline{w})$ is convex, we want to show that $w_i, i=0, 1, \dots, M$, solving $\frac{\partial E}{\partial w_i} = 0$,

(i.e., minimizing $E(\underline{w})$) solve/satisfy

$$\sum_{j=0}^M \sum_{n=1}^N (x_n)^{i+j} w_j = \sum_{n=1}^N (x_n)^i t_n.$$

To this end, we simply note that

$$\frac{\partial E}{\partial w_i} = \frac{\partial}{\partial w_i} \left[\frac{1}{2} \sum_{n=1}^N (y(x_n, \underline{w}) - t_n)^2 \right]$$

$$= \frac{1}{2} \sum_{n=1}^N \frac{\partial}{\partial w_i} (y(x_n, \underline{w}) - t_n)^2$$

$$\stackrel{\text{Chain rule}}{\Rightarrow} \frac{1}{2} \sum_{n=1}^N 2(y(x_n, \underline{w}) - t_n) \frac{\partial}{\partial w_i} y(x_n, \underline{w})$$

$$= \frac{1}{2} \sum_{n=1}^N 2(y(x_n, \underline{w}) - t_n) x_n^i \quad \swarrow \text{look at def. of } y(x_n, \underline{w})$$

$$= \sum_{n=1}^N \left(\left(\sum_{j=0}^M w_j x_n^j \right) - t_n \right) x_n^i$$

$$= \sum_{n=1}^N \left(\sum_{j=0}^M w_j x_n^j \right) x_n^i - t_n x_n^i$$

Now, consider

$$\sum_{n=1}^N \left(\sum_{j=0}^M w_j x_n^j \right) x_n^i - t_n x_n^i = 0$$

$$\Rightarrow \sum_{n=1}^N \sum_{j=0}^M w_j x_n^j x_n^i = \sum_{n=1}^N t_n x_n^i$$

$$\Rightarrow \sum_{n=1}^N \sum_{j=0}^M w_j x_n^{(j+i)} = \sum_{n=1}^N t_n x_n^i$$

as we wanted.

Book exercise 1.12)

Let $x_n, x_m \sim U(\mu, \sigma^2)$ be i.i.d. and

$$I_{n,m} = \begin{cases} 1, & \text{if } n=m \\ 0, & \text{else} \end{cases}.$$

We want to show $E[x_n x_m] = \mu^2 + I_{n,m} \sigma^2$.

To this end, recall the following results from the last week's class (some might be redundant).

Let x, y be random variables and $a, b \in \mathbb{R}$ be constants. Then

(1) $\text{Var}(x) = E[x^2] - E[x]^2$,

(2) $\text{Var}(a+bx) = b^2 \text{Var}(x)$

(3) $\text{Var}(x+y) = \text{Var}(x) + \text{Var}(y) \pm 2\text{Cov}(x, y)$

(4) $\text{Cov}(x, y) = E[xy] - E[x]E[y]$

(5) $x \perp y \Rightarrow \text{Cov}(x, y) = 0$.

Now, if $n=m$, we immediately have

$$E[x_n^2] = E[x_m^2] = \mu^2 + \sigma^2$$

from (1).

For $n \neq m$, it follows from (4) and (5) that

$$E[x_n x_m] = \mu^2.$$

Thus, we have $E[x_n x_m] = \mu^2 + I_{n,m} \sigma^2$.

We now seek to show (1.57) and (1.58), i.e.,

$$E[\mu_{ML}] = \mu \quad \text{where} \quad \mu_{ML} = \frac{1}{N} \sum_{n=1}^N x_n,$$

and

$$E[\sigma_{ML}^2] = \left(\frac{N-1}{N}\right) \sigma^2 \quad \text{where} \quad \sigma_{ML}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{ML})^2.$$

To this end, we first note that

$$E[\mu_{ML}] = E\left[\frac{1}{N} \sum_{n=1}^N x_n\right] = \frac{1}{N} \sum_{n=1}^N E[x_n] = \frac{1}{N} \cdot N \cdot \mu = \mu.$$

Furthermore,

$$\begin{aligned} E[\sigma_{ML}^2] &= E\left[\frac{1}{N} \sum_{n=1}^N (x_n - \mu_{ML})^2\right] \\ &= \frac{1}{N} E\left[\sum_{n=1}^N x_n^2 + \mu_{ML}^2 - 2x_n \mu_{ML}\right] \\ &= \frac{1}{N} \sum_{n=1}^N E[x_n^2] + \frac{1}{N} \sum_{n=1}^N E[\mu_{ML}^2] - \frac{2}{N} E\left[\sum_{n=1}^N x_n \mu_{ML}\right] \end{aligned}$$

by start
of exercise

$$= \mu^2 + \sigma^2 + E\left[\left(\frac{1}{N} \sum_{m=1}^N x_m\right)^2\right] - \frac{2}{N} E\left[\sum_{n=1}^N x_n \left(\frac{1}{N} \sum_{m=1}^N x_m\right)\right]$$

$$= \mu^2 + \sigma^2 + \frac{1}{N^2} E\left[\left(\sum_{m=1}^N x_m\right)^2\right] - \frac{2}{N^2} E\left[\left(\sum_{m=1}^N x_m\right) \left(\sum_{n=1}^N x_n\right)\right]$$

$$= \mu^2 + \sigma^2 - \frac{1}{N^2} E\left[\left(\sum_{m=1}^N x_m\right) \left(\sum_{n=1}^N x_n\right)\right]$$

$$= \mu^2 + \sigma^2 - \frac{1}{N^2} E\left[\sum_{m=1}^N \sum_{n=1}^N x_m x_n\right]$$

$$= \mu^2 + \sigma^2 - \frac{1}{N^2} E\left[\sum_{m=1}^N \sum_{n=1}^N \mathbb{1}_{\{m=n\}} x_m x_n\right]$$

$$\begin{array}{l} \text{\# elements} \swarrow \quad \searrow \text{\# elements} \\ -\frac{1}{N^2} E\left[\left(\sum_{m=1}^N \sum_{n=1}^N \mathbb{1}_{\{m \neq n\}} x_m x_n\right)\right] \end{array}$$

$$= \mu^2 + \sigma^2 - \frac{1}{N^2} N \cdot (\mu^2 + \sigma^2) - \frac{1}{N^2} (N \cdot (N-1)) \cdot (\mu^2)$$

$$= \mu^2 + \sigma^2 - \frac{1}{N} (\mu^2 + \sigma^2) - \frac{1}{N} (N-1) \mu^2$$

$$= \mu^2 + \sigma^2 - \frac{1}{N} (\mu^2 + \sigma^2) - \frac{1}{N} (N\mu^2 - \mu^2)$$

$$= \mu^2 + \sigma^2 - \frac{1}{N} \mu^2 - \frac{1}{N} \sigma^2 - \mu^2 + \frac{1}{N} \mu^2$$

$$= \mu^2 + \sigma^2 - \frac{1}{N} \sigma^2 - \mu^2$$

$$= \sigma^2 - \frac{1}{N} \sigma^2$$

$$= \frac{N\sigma^2}{N} - \frac{1}{N} \sigma^2$$

$$= \frac{(N-1)}{N} \sigma^2.$$

Book exercise 1.22)

The loss matrix can be written as

$$L = \mathbf{1} - \mathbf{I} = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{matrix} \text{True class} \uparrow \\ k \end{matrix}$$

Predicted
class $j \rightarrow$

If we assign/predict the right class for an observation $\underline{x}$, we incur no loss. Else we incur a loss with a factor of 1 (no matter what misclassification we made).

As stated in the book, the expected risk

$$\mathbb{E}[L] = \sum_k \sum_j \int L_{k,j} p(\underline{x} | \mathcal{C}_k) d\underline{x}$$

is minimized if, for each $\underline{x}$, we choose/predict the class j that minimizes (1.81), i.e.,

$$\begin{aligned} \sum_k L_{k,j} p(\mathcal{C}_k | \underline{x}) &= \sum_k (1 - I_{k,j}) p(\mathcal{C}_k | \underline{x}) \\ &= \underbrace{\sum_k p(\mathcal{C}_k | \underline{x})}_{=1} - p(\mathcal{C}_j | \underline{x}) \end{aligned}$$

which, indeed, is minimized by

$$j = \arg \max_q p(\mathcal{C}_q | \underline{x}),$$

i.e., predicting the class with the largest posterior probability.

Last extra exercise)

We will consider the figure given on the exercise.

The statement " $p(C_1) = 2p(C_2)$ " is interpreted to mean that the prior probability of class C_1 is two times as big as the prior probability of class C_2 .

We specifically have two equations in two unknowns

$$p(C_1) = 2p(C_2),$$

$$p(C_1) + p(C_2) = 1$$

$$\Rightarrow p(C_1) = \frac{2}{3}, p(C_2) = \frac{1}{3}.$$

(1) To find $p(x < 6)$, we use the law of total probability,

$$p(x < 6) = \sum_{k=1}^2 p(x < 6 | C_k) \cdot p(C_k)$$

$$= \overset{\substack{f+c+a+e+g \\ \text{must sum to 1}}}{1} \cdot p(C_1) + (h+d) \cdot p(C_2)$$

$$= 1 \cdot \frac{2}{3} + (0.04 + 0.20) \cdot \frac{1}{3}$$

$$= \frac{56}{75}$$

$$\approx 0.7466\%$$

(2) It is hopefully obvious that

$$p(x < 6 | C_1) = 1.$$

Formally, you could write up something like

$$\begin{aligned} p(x < 6 | C_1) &= p(x < 1 | C_1) + p(x \in [1, 2] | C_1) \\ &\quad + p(x \in [2, 3] | C_1) + \dots + p(x \in [5, 6] | C_1) + p(x > 6 | C_1) \\ &= 0 + f + c + a + e + g + 0 \\ &= 1 \end{aligned}$$

(but this is really not necessary(!)).

(3) We apply the following strategy, seeking to minimize the expected loss.

First, note by Bayes theorem

$$p(C_1 | x = 4.5) = \frac{p(x = 4.5 | C_1) \cdot p(C_1)}{p(x = 4.5)}, \text{ and}$$

$$p(C_2 | x = 4.5) = \frac{p(x = 4.5 | C_2) \cdot p(C_2)}{p(x = 4.5)}.$$

Notice how both terms on the right-hand side are divided by " $p(x = 4.5)$ ". Because we

are interested in the proportion between $p(C_1|x=4.5)$ and $p(C_2|x=4.5)$, we shall ignore this denominator.

From the figure on the exercise, we read that

$$p(x=4.5|C_1) = e = 0.17, \text{ and}$$

$$p(x=4.5|C_2) = h = 0.04.$$

Note: Problems with pointwise probabilities of continuous random variables.

Now

Means "proportional to"

$$p(C_1|x=4.5) \propto p(x=4.5|C_1) \cdot p(C_1) = 0.17 \cdot \frac{2}{3} = \frac{17}{150},$$

$$p(C_2|x=4.5) \propto p(x=4.5|C_2) \cdot p(C_2) = 0.04 \cdot \frac{1}{3} = \frac{1}{75}.$$

So without considering the loss matrix, we would assign $x=4.5$ to class C_2 .

The loss matrix tells us that misclassifying an instance from class C_1 is twice as "costly" as misclassifying an instance of class C_2 .

However, we still have that

$$\frac{1}{2} \cdot p(C_1|x=4.5) \propto \frac{1}{2} \cdot \frac{17}{150} > \frac{1}{75} = p(C_2|x=4.5)$$

and still assign $x=4.5$ to class C_1 , even when we account of the loss.